INCH - POUND

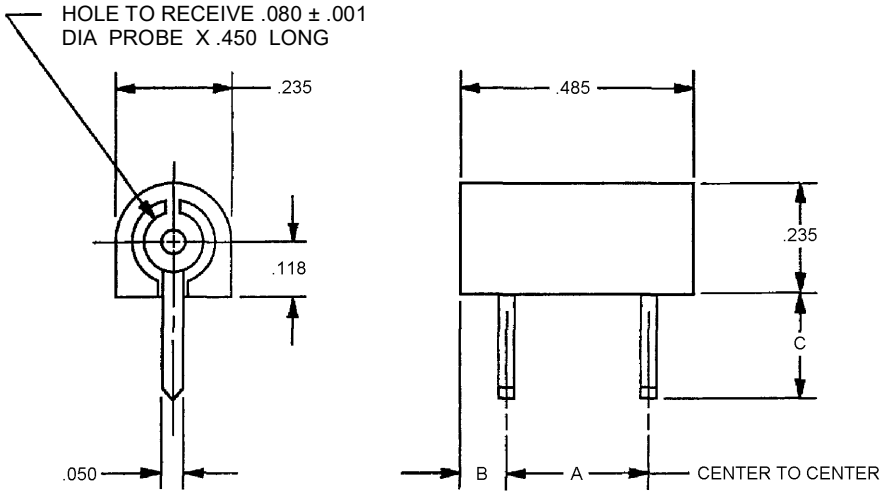
MIL-DTL-39024/13D
w/AMENDMENT 2
3 June 2020
SUPERSEDING
MIL-DTL-39024/13D
w/AMENDMENT 1
01 December 2018

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, TEST POINT TYPE, PRINTED WIRING TYPE; SINGLE TEST POINT (RIGHT ANGLE, 2-LEG MOUNTING), LOW VOLTAGE, .080

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-39024.



Inches	mm
.001	0.03
.050	1.27
.080	2.03
.118	3.00
.235	5.97
.485	12.32
.5	12.7

NOTES:
1. Dimensions are in inches. Metric equivalents are given for information only.
2. See table I for dimensions not specified on figure 1.
3. Unless otherwise specified, tolerances are ± .010 (0.25 mm), and are based on 1 inch = 25.4 mm.
4. All undimensioned pictorial configurations are for reference purposes only.

FIGURE 1. Configuration and dimensions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:26Z
Check the source to verify that this is the current version before use.

TABLE I. Design and identification data.

Dash number	Dimensions (inches) 1/			Insulation
	A ± .010 (0.25)	B ± .015 (0.38)	C ± .015 (0.38)	Color in accordance with SAE-AMS-STD-595
-01	.300 (7.62)	0.060 (1.52)	0.218 (5.54)	White
-02	"	"	"	Red
-03	"	"	"	Black
-04	"	"	"	Brown
-05	"	"	"	Green
-06	"	"	"	Orange
-07	"	"	"	Blue
-08	"	"	"	Yellow
-09	"	"	"	Gray
-10	"	"	"	Purple
-11	"	"	"	Natural
-12	.400 (10.16)	.025 (0.64)	.175 (4.45)	White
-13	"	"	"	Red
-14	"	"	"	Black
-15	"	"	"	Brown
-16	"	"	"	Green
-17	"	"	"	Orange
-18	"	"	"	Blue
-19	"	"	"	Yellow
-20	"	"	"	Gray
-21	"	"	"	Purple
-22	"	"	"	Natural

1/ Dimensions are in inches. Metric equivalents are given for information only.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1 and table I.

Insulation: Material shall be polyamide in accordance with L-P-410 or ASTM D4066.

Flammability: Dielectric materials shall be self-extinguishing in accordance with ASTM D635.

Test probe: 0.080 ± 0.001-inch diameter and .450 inch (min) length. Test probe shall be engaged a minimum of .450 inches. Test probe shall be fully inserted.

Contact current rating: 5 amperes (maximum).

Contact sleeve: Shall be formed of strip to make a cylinder with a butted seam.

Contact resistance: A test probe in accordance with MIL-DTL-39024 shall be fully inserted in the connector. With a current of 5 amperes, potential drop shall be measured between the connector terminal and the extreme end of the solder cup on the test probe. Measured values shall be within the limits specified in table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:26Z
Check the source to verify that this is the current version before use.

TABLE II. Potential drop (max) with 5 amperes of current.

Test	Before test	After test
Durability	6 mV	8 mV
Vibration	8 mV	10 mV
Salt spray (corrosion)	10 mV	15 mV

Dielectric withstanding voltage (at sea level):

Test voltage: 3,800 volts rms, 60 hertz, shall be applied for a period of 15 seconds.

Insertion and withdrawal forces:

Insertion force: 5.0 pounds (max).
Withdrawal force: 0.4 pound (min).

Identification marking:

Part or Identifying Number (PIN): M39024/13-(dash number from table I).

Operating conditions:

Operating voltage: 2,400 volts rms, 60 hertz at sea level; 350 volts rms, 60 hertz at 80,000 feet.

Qualification: Not applicable.

First article inspection. First article inspection shall consist of Group A and B inspections. MIL-DTL-39024 paragraph 4.5 through 4.5.3.1.3 and Tables III through VII as applicable.

Conformance inspection. Inspection of successive lots for delivery shall consist of Group A inspections. MIL-DTL-39024 paragraph 4.5 through 4.5.3.1.3 and Tables III through VII as applicable.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-39024, this document references the following:

ASTM D4066
ASTM D635
SAE-AMS-STD-595
L-P-410

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:26Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2020-081)

Review activities:
Army - AR, CR4, MI
Navy - AS, CG, MC, OS
Air Force - 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:26Z
Check the source to verify that this is the current version before use.